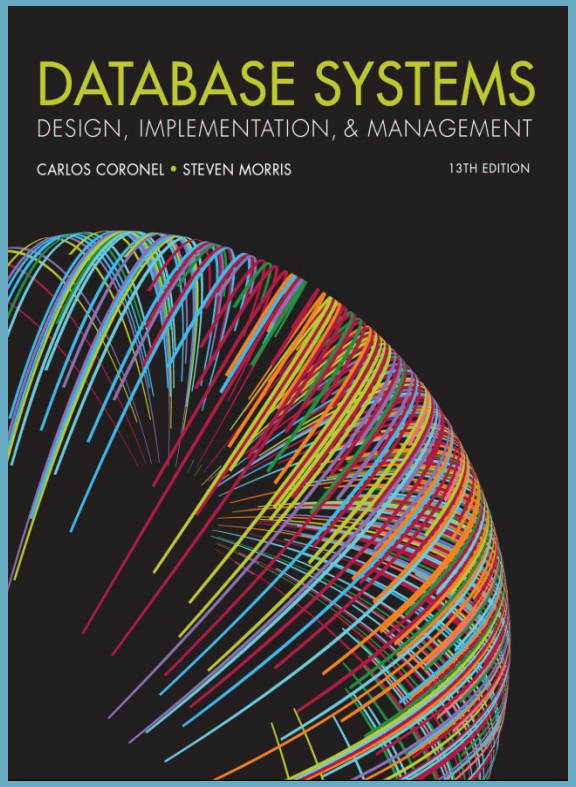


Chapter 3

The Relational Database Model

Learning Objectives

- After completing this chapter, you will be able to:
  - Describe the relational database model's logical structure
  - Identify the relational model's basic components and explain the structure, contents, and characteristics of a relational table
  - Use relational database operators to manipulate relational table contents
  - Explain the purpose and components of the data dictionary and system catalog
  - Identify appropriate entities and then the relationships among the entities in the relational database model
  - Describe how data redundancy is handled in the relational database model
  - Explain the purpose of indexing in a relational database

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The Entity Relationship Model

- Entity relationship diagram (ERD)
  - Uses graphic representations to model database components
  - Entity is mapped to a relational table

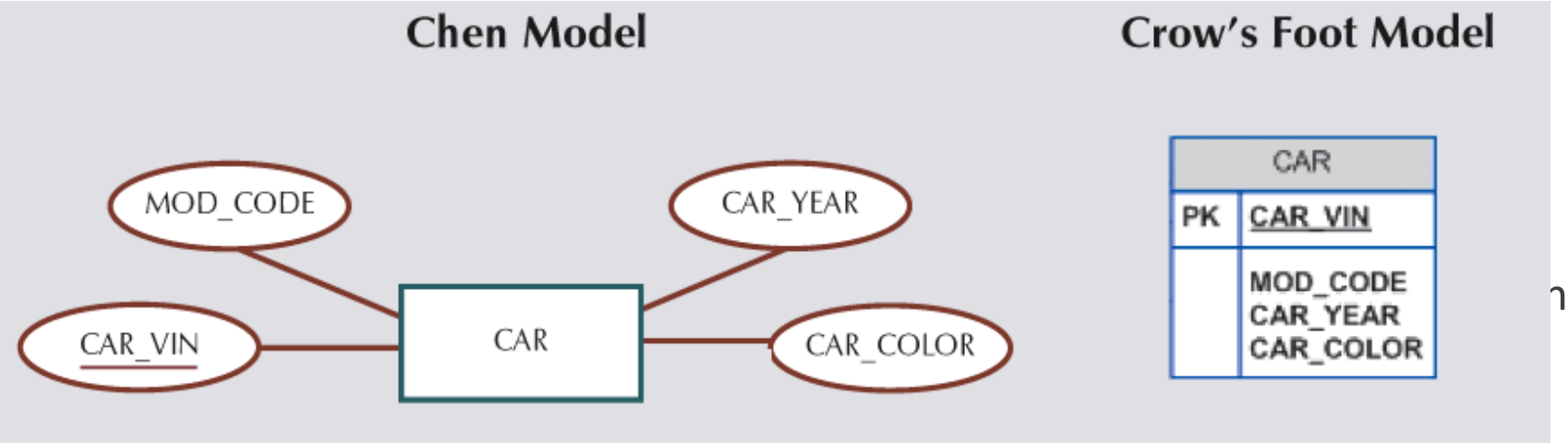
- Entities – are objects/concepts identified by the enterprise as having an independent existence
- Relationships – are association of entities where the association includes one entity from each participating entity type

Basic Rules for Modelling Entities

- Each entity should have multiple occurrences
  - Entity instance (or occurrence) is row in table
  - Entity set is collection of like entities
- Each entity must have at least one descriptive attribute (e.g. Student_Name, Student_DOB). The number of attributes refers to the degree of the entity.
- Entities must not be free-standing (i.e. must participate in at least one relationship).
- Entities must have some unique identifier that helps to identify a particular occurrence of an entity (or primary key)

Modelling Entities

- Chen's notation and Crow's notation are used to model entities differently.



- H
th

Basic Rules for Modelling Relationships

- **Participants** are entities that participate in a relationship
- Relationships between entities always operate in both directions
- **Connectivity** - Describes the relationship classification
- **Cardinality** - Expresses minimum and maximum number of entity occurrences associated with one occurrence of related entity
- Relationships are established by very concise statements known as business rules

Modelling Relationships

- Relationships expressed using Chen notation
  - Relationships represented by a diamond
  - Relationship name written inside the diamond



- Relationships expressed using Crow notation
  - Relationships represented by a line between the two entities
  - Relationship name written over the line



Relationships

- Connectivity labels type of relationships. There are 3 types of relationships:
  - One-to-one (1:1) : An academic department is chaired by one professor; a professor may chair only one academic department.
  - one-to-many (1:M) : A customer may generate many invoices; each invoice is generated by one customer.
  - many-to-many (M:N or M:M.) : An employee may have earned many degrees; a degree may have been earned by many employees.

- Relationships are read from the ‘1’ side to the ‘M’ side

A Logical View of Data

- Relational database model enables logical representation of the data and its relationships
  - Logical simplicity yields simple and effective database design methodologies
  - The logical view is facilitated by the creation of data relationships based on a logical construct called a relation

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Tables and Their Characteristics

Table 3.1	Characteristics of a Relational Table
1	A table is perceived as a two-dimensional structure composed of rows and columns.
2	Each table row (tuple) represents a single entity occurrence within the entity set.
3	Each table column represents an attribute, and each column has a distinct name.
4	Each intersection of a row and column represents a single data value.
5	All values in a column must conform to the same data format.
6	Each column has a specific range of values known as the attribute domain.
7	The order of the rows and columns is immaterial to the DBMS.
8	Each table must have an attribute or combination of attributes that uniquely identifies each row.

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Keys

- Keys consist of one or more attributes that determine other attributes
  - Ensure that each row in a table is uniquely identifiable
  - Establish relationships among tables and to ensure the integrity of the data
- Primary key: attribute or combination of attributes that uniquely identifies a row

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Dependencies

- Determination
  - State in which knowing the value of one attribute makes it possible to determine the value of another
    - Establishes the role of a key
    - Based on the relationships among the attributes
- Functional dependence: value of one or more attributes determines the value of one or more other attributes
  - Determinant: attribute whose value determines another
  - Dependent: attribute whose value is determined by the other attribute
- Full functional dependence: entire collection of attributes in the determinant is necessary for the relationship

A determines B

A → B

EMP_NUM → EMP_LNAME, EMP_INITIAL, EMP_FNAME

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Types of Keys (1 of 3)

- Several different types of keys are used in the relational model
  - Composite key: key that is composed of more than one attribute
  - Key attribute: attribute that is a part of a key
  - Superkey: key that can uniquely identify any row in the table
  - Candidate key: minimal superkey
  - Entity integrity: condition in which each row in the table has its own unique identity
    - All of the values in the primary key must be unique
    - No key attribute in the primary key can contain a null
  - Null: absence of any data value
    - Unknown attribute value, known but missing attribute value, or inapplicable condition
  - Referential integrity: every reference to an entity instance by another entity instance is valid
  - Foreign key: primary key of one table that has been placed into another table to create a common attribute
  - Secondary key: key used strictly for data retrieval purposes

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Types of Keys (2 of 3)

FIGURE 3.2 AN EXAMPLE OF A SIMPLE RELATIONAL DATABASE

Table name: PRODUCT
Primary key: PROD_CODE
Foreign key: VEND_CODE

Database name: Ch03_SaleCo

PROD_CODE	PROD_DESCRIPT	PROD_PRICE	PROD_ON_HAND	VEND_CODE
001278-AB	Claw hammer	12.95	23	232
123-21UUY	Houselite chain saw, 16-in. bar	189.99	4	235
QER-34256	Sledge hammer, 16-lb. head	18.63	6	231
SRE-657UG	Rat-tail file	2.99	15	232
ZZX/3245Q	Steel tape, 12-ft. length	6.79	8	235

link

Table name: VENDOR
Primary key: VEND_CODE
Foreign key: none

VEND_CODE	VEND_CONTACT	VEND_AREACODE	VEND_PHONE
230	Shelly K. Smithson	608	555-1234
231	James Johnson	615	123-4536
232	Annelise Crystall	608	224-2134
233	Candice Wallace	904	342-6567
234	Arthur Jones	615	123-3324
235	Henry Ortozo	615	899-3425

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Types of Keys (3 of 3)

Table 3.3	Relational Database Keys
Key Type	**Definition**
Superkey	An attribute or combination of attributes that uniquely identifies each row in a table.
Candidate key	A minimal (irreducible) superkey; a superkey that does not contain a subset of attributes that is itself a superkey.
Primary key	A candidate key selected to uniquely identify all other attribute values in any given row; cannot contain null entries.
Foreign key	An attribute or combination of attributes in one table whose values must either match the primary key in another table or be null.
Secondary key	An attribute or combination of attributes used strictly for data retrieval purposes.

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Does the following specific classification represent a superkey scenario?

TABLE 3.2 Student Classification

HOURS COMPLETED	CLASSIFICATION
Less than 30	Fr
30–59	So
60–89	Jr
90 or more	Sr

NO – because the specific number of hours is not dependent on the classification

i.e. CLASSIFICATION does not determine one and only one value for HOURS COMPLETED.

Integrity Rules (1 of 4)

- Relational database integrity rules are very important to good database design
  - Relational database management systems (RDBMSs) enforce integrity rules automatically
    - Much safer to make sure the application design conforms to entity and referential integrity rules

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Integrity Rules (2 of 4)

Table 3.4	Integrity Rules
Entity Integrity	**Description**
Requirement	All primary key entries are unique, and no part of a primary key may be null.
Purpose	Each row will have a unique identity, and foreign key values can properly reference primary key values.
Example	No invoice can have a duplicate number, nor can it be null; in short, all invoices are uniquely identified by their invoice number.
Referential Integrity	**Description**
Requirement	A foreign key may have either a null entry, as long as it is not a part of its table's primary key, or an entry that matches the primary key value in a table to which it is related (every non-null foreign key value must reference an existing primary key value).
Purpose	It is possible for an attribute not to have a corresponding value, but it will be impossible to have an invalid entry; the enforcement of the referential integrity rule makes it impossible to delete a row in one table whose primary key has mandatory matching foreign key values in another table.
Example	A customer might not yet have an assigned sales representative (number), but it will be impossible to have an invalid sales representative (number).

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Integrity Rules (3 of 4)

FIGURE 3.3 AN ILLUSTRATION OF INTEGRITY RULES

Table name: CUSTOMER **Database name: Ch03_InsureCo**
Primary key: CUS_CODE
Foreign key: AGENT_CODE

CUS_CODE	CUS_LNAME	CUS_FNAME	CUS_INITIAL	CUS_RENEW_DATE	AGENT_CODE
10010	Ramas	Alfred	A	05-Apr-2018	502
10011	Dunne	Leona	K	16-Jun-2018	501
10012	Smith	Kathy	W	29-Jan-2019	502
10013	Olowski	Paul	F	14-Oct-2018	
10014	Orlando	Myron		28-Dec-2018	501
10015	O'Brian	Amy	B	22-Sep-2018	503
10016	Brown	James	G	25-Mar-2019	502
10017	Williams	George		17-Jul-2018	503
10018	Farriss	Anne	G	03-Dec-2018	501
10019	Smith	Olette	K	14-Mar-2019	503

Table name: AGENT (only five selected fields are shown)
Primary key: AGENT_CODE
Foreign key: none

AGENT_CODE	AGENT_AREACODE	AGENT_PHONE	AGENT_LNAME	AGENT_YTD_SLS
501	713	228-1249	Alby	132735.75
502	615	882-1244	Hahn	138967.35
503	615	123-5589	Okon	127093.45

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Integrity Rules (4 of 4)

- **Null**: Absence of any data value that could represent:
  - An unknown attribute value
  - A known, but missing, attribute value
  - A “not applicable” condition
  - Can create problems when functions such as COUNT, AVERAGE, and SUM are used
  - Can create logical problems when relational tables are linked
- Ways to handle nulls
  - Flags
    - Special codes used to indicate the absence of some value
  - Constraints
    - NOT NULL constraint: placed on a column to ensure that every row in the table has a value for that column
    - UNIQUE constraint: restriction placed on a column to ensure that no duplicate values exist for that column

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Integrity Constraints

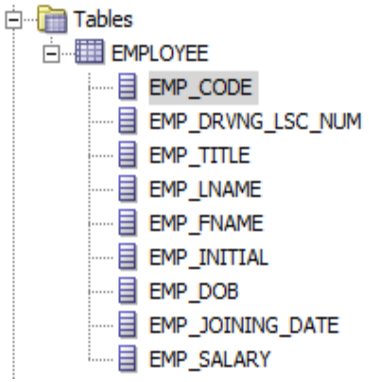
- When creating tables, you typically create them with integrity constraints. These constraints enforce the business rules of a system. They ease the burden of programming the business rules in individual applications as the database enforces the constraint.

- Some constraints are – PRIMARY KEY, UNIQUE, NOT NULL, CHECK, CONSTRAINT
- These constraints can be applied to individual attributes on each row or as named constraints)

Integrity Constraints

- Some examples where we apply constraints/rules - employee salary cannot be negative, employee name cannot be null, set joining date to a default date, license number must be unique

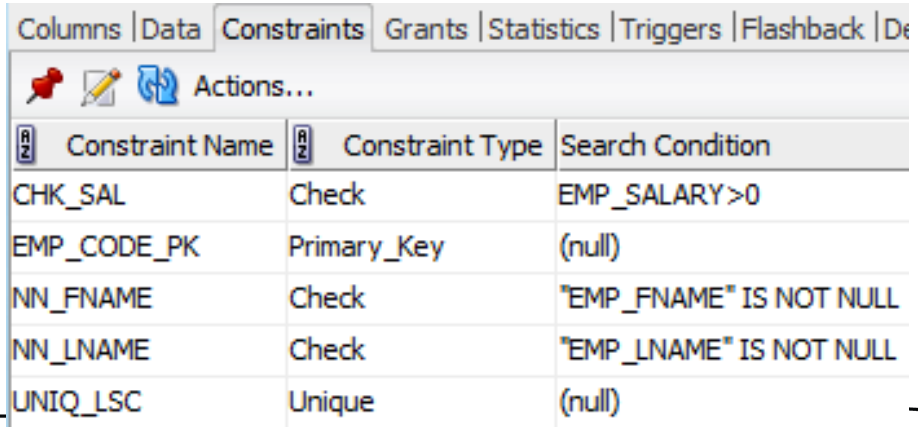
```
CREATE TABLE EMPLOYEE (
EMP_CODE NUMBER(10) PRIMARY KEY,
EMP_DRVNG_LSC_NUM VARCHAR2(20) UNIQUE,
EMP_TITLE VARCHAR2(4),
EMP_LNAME VARCHAR2(15) NOT NULL,
EMP_FNAME VARCHAR2(15) NOT NULL,
EMP_INITIAL VARCHAR2(1),
EMP_DOB DATE,
EMP_JOINING_DATE DATE DEFAULT SYSDATE,
EMP_SALARY NUMBER(10) CHECK(EMP_SALARY > 0)
);
```



Constraint Name	Constraint Type	Search Condition
SYS_C0011568	Check	"EMP_LNAME" IS NOT NULL
SYS_C0011569	Check	"EMP_FNAME" IS NOT NULL
SYS_C0011570	Check	EMP_SALARY > 0
SYS_C0011571	Primary_Key	(null)
SYS_C0011572	Unique	(null)

Integrity Constraints

```
CREATE TABLE EMPLOYEE1 (
EMP_CODE NUMBER(10) CONSTRAINT EMP_CODE_PK PRIMARY KEY,
EMP_DRVNG_LSC_NUM VARCHAR2(20) CONSTRAINT UNIQ_LSC UNIQUE,
EMP_TITLE VARCHAR2(4),
EMP_LNAME VARCHAR2(15) CONSTRAINT NN_LNAME NOT NULL,
EMP_FNAME VARCHAR2(15) CONSTRAINT NN_FNAME NOT NULL,
EMP_INITIAL VARCHAR2(1),
EMP_DOB DATE,
EMP_JOINING_DATE DATE DEFAULT SYSDATE,
EMP_SALARY NUMBER(10) CONSTRAINT CHK_SAL CHECK(EMP_SALARY>0)
);
```





Other Constraints

Foreign Key…

```
CREATE TABLE DEPT(DEPT_ID NUMBER (3) PRIMARY KEY,
        DEPT_NAME VARCHAR2(20));

CREATE TABLE EMP(EMP_ID NUMBER(3), EMP_NAME VARCHAR2(10),
        DEPT_ID NUMBER(3) REFERENCES DEPT(DEPT_ID));
```

Named constraints…

```
CREATE TABLE DEPT(DEPT_ID NUMBER (3), DEPT_NAME VARCHAR2(20),
        CONSTRAINT DEPT_ID_PK PRIMARY KEY(DEPT_ID));
```

Examples

```
QUANTITY NUMBER CONSTRAINT NN_QTY NOT NULL
CONSTRAINT CHECK_QTY CHECK (QUANTITY > 0),
COST NUMBER CONSTRAINT CHECK_COST CHECK (cost > 0)
```

Example

- An EMPLOYEE has only one JOB_CODE, but a JOB_CODE can be held by many EMPLOYEEs.
- An EMPLOYEE can participate in many PLANs, and any PLAN can be assigned to many EMPLOYEEs.

Questions

1. Identify the primary key and the foreign key(s)
2. Which tables exhibit entity integrity or referential integrity?
3. Draw an ERD diagram

Table name: EMPLOYEE

EMP_CODE	EMP_LNAME	JOB_CODE
14	Rudell	2
15	McDade	1
16	Ruellardo	1
17	Smith	3
20	Smith	2

Table name: BENEFIT

EMP_CODE	PLAN_CODE
15	2
15	3
16	1
17	1
17	3
17	4
20	3

Table name: JOB

JOB_CODE	JOB_DESCRIPTION
1	Clerical
2	Technical
3	Managerial

Table name: PLAN

PLAN_CODE	PLAN_DESCRIPTION
1	Term life
2	Stock purchase
3	Long-term disability
4	Dental



Relational Algebra

- Theoretical way of manipulating table contents using relational operators
  - Relvar: variable that holds a relation
    - Heading contains the names of the attributes
    - Body contains the relation
  - Relational operators have the property of closure
    - Closure: use of relational algebra operators on existing relations produces new relations

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Relational Set Operators (1 of 10)

- Select (restrict)
  - Unary operator that yields a horizontal subset of a table
- Project
  - Unary operator that yields a vertical subset of a table
- Union
  - Combines all rows from two tables, excluding duplicate rows
  - Union-compatible: tables share the same number of columns, and their corresponding columns share compatible domains
- Intersect
  - Yields only the rows that appear in both tables
  - Tables must be union-compatible to yield valid results

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Relational Set Operators (2 of 10)

FIGURE 3.4 SELECT

Original table

P_CODE	P_DESCRIPT	PRICE
123456	Flashlight	5.26
123457	Lamp	25.15
123458	Box Fan	10.99
213345	9v battery	1.92
254467	100W bulb	1.47
311452	Powerdrill	34.99

SELECT ALL yields

New table

P_CODE	P_DESCRIPT	PRICE
123456	Flashlight	5.26
123457	Lamp	25.15
123458	Box Fan	10.99
213345	9v battery	1.92
254467	100W bulb	1.47
311452	Powerdrill	34.99

SELECT only PRICE less than $2.00 yields

P_CODE	P_DESCRIPT	PRICE
213345	9v battery	1.92
254467	100W bulb	1.47

SELECT only P_CODE = 311452 yields

P_CODE	P_DESCRIPT	PRICE
311452	Powerdrill	34.99

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Relational Set Operators (3 of 10)

FIGURE 3.5 PROJECT

Original table

P_CODE	P_DESCRIPT	PRICE
123456	Flashlight	5.26
123457	Lamp	25.15
123458	Box Fan	10.99
213345	9v battery	1.92
254467	100W bulb	1.47
311452	Powerdrill	34.99

PROJECT PRICE yields

New table

PRICE
5.26
25.15
10.99
1.92
1.47
34.99

PROJECT P_DESCRIPT and PRICE yields

P_DESCRIPT	PRICE
Flashlight	5.26
Lamp	25.15
Box Fan	10.99
9v battery	1.92
100W bulb	1.47
Powerdrill	34.99

PROJECT P_CODE and PRICE yields

P_CODE	PRICE
123456	5.26
123457	25.15
123458	10.99
213345	1.92
254467	1.47
311452	34.99

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Relational Set Operators (4 of 10)

FIGURE 3.6 UNION

P_CODE	P_DESCRIPT	PRICE
123456	Flashlight	5.26
123457	Lamp	25.15
123458	Box Fan	10.99
213345	9v battery	1.92
254467	100W bulb	1.47
311452	Powerdrill	34.99

UNION

P_CODE	P_DESCRIPT	PRICE
345678	Microwave	160.00
345679	Dishwasher	500.00
123458	Box Fan	10.99

yields

P_CODE	P_DESCRIPT	PRICE
123456	Flashlight	5.26
123457	Lamp	25.15
123458	Box Fan	10.99
213345	9v battery	1.92
254467	100W bulb	1.47
311452	Powerdrill	34.99
345678	Microwave	160
345679	Dishwasher	500

FIGURE 3.7 INTERSECT

STU_FNAME	STU_LNAME
George	Jones
Jane	Smith
Peter	Robinson
Franklin	Johnson
Martin	Lopez

INTERSECT

EMP_FNAME	EMP_LNAME
Franklin	Lopez
William	Turner
Franklin	Johnson
Susan	Rogers

yields

STU_FNAME	STU_LNAME
Franklin	Johnson

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Relational Set Operators (5 of 10)

- Difference
  - Yields all rows in one table that are not found in the other table
  - Tables must be union-compatible to yield valid results

FIGURE 3.8 DIFFERENCE

STU_FNAME	STU_LNAME
George	Jones
Jane	Smith
Peter	Robinson
Franklin	Johnson
Martin	Lopez

DIFFERENCE

EMP_FNAME	EMP_LNAME
Franklin	Lopez
William	Turner
Franklin	Johnson
Susan	Rogers

yields

STU_FNAME	STU_LNAME
George	Jones
Jane	Smith
Peter	Robinson
Martin	Lopez

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Relational Set Operators (6 of 10)

- Product
  - Yields all possible pairs of rows from two tables

FIGURE 3.9 PRODUCT

P_CODE	P_DESCRIPT	PRICE
123456	Flashlight	5.26
123457	Lamp	25.15
123458	Box Fan	10.99
213345	9v battery	1.92
254467	100W bulb	1.47
311452	Powerdrill	34.99

PRODUCT

STORE	AISLE	SHELF
23	W	5
24	K	9
25	Z	6

yields

P_CODE	P_DESCRIPT	PRICE	STORE	AISLE	SHELF
123456	Flashlight	5.26	23	W	5
123456	Flashlight	5.26	24	K	9
123456	Flashlight	5.26	25	Z	6
123457	Lamp	25.15	23	W	5
123457	Lamp	25.15	24	K	9
123457	Lamp	25.15	25	Z	6
123458	Box Fan	10.99	23	W	5
123458	Box Fan	10.99	24	K	9
123458	Box Fan	10.99	25	Z	6
213345	9v battery	1.92	23	W	5
213345	9v battery	1.92	24	K	9
213345	9v battery	1.92	25	Z	6
311452	Powerdrill	34.99	23	W	5
311452	Powerdrill	34.99	24	K	9
311452	Powerdrill	34.99	25	Z	6
254467	100W bulb	1.47	23	W	5
254467	100W bulb	1.47	24	K	9
254467	100W bulb	1.47	25	Z	6

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Relational Set Operators (7 of 10)

- Joins allow information to be intelligently combined from two or more tables
  - Natural join: links tables by selecting only the rows with common values in their common attribute
  - Equijoin: links tables on the basis of an equality condition that compares specified columns of each table
  - Theta join: links tables using an inequality comparison operator
  - Inner join: only returns matched records from the tables that are being joined
  - Outer join: matched pairs are retained and unmatched values in the other table are left null
    - Left outer join: yields all of the rows in the first table, including those that do not have a matching value in the second table
    - Right outer join: yields all of the rows in the second table, including those that do not have matching values in the first table

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Relational Set Operators (8 of 10)

- Divide
  - Uses one double-column table as the dividend and one single-column table as the divisor
  - Output is a single column that contains all values from the second column of the dividend that are associated with every row in the divisor

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Relational Set Operators (10 of 10)

FIGURE 3.16 DIVIDE

P_CODE	CUS_CODE
123456	10400
123456	11501
123456	10030
123456	12550
234567	12350
234567	10040
234567	10500
234567	10030
234567	12550
345678	10400
345678	11630
345678	12550
456789	11630
567890	10500
567890	10030
567890	12550
678901	11500
678901	10400
678901	11630

DIVIDE

P_CODE
123456
234567
567890

yields

CUS_CODE
10030
12550

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Relational Set Operators (9 of 10)

FIGURE 3.10 TWO TABLES THAT WILL BE USED IN JOIN ILLUSTRATIONS

Table name: CUSTOMER

CUS_CODE	CUS_LNAME	CUS_ZIP	AGENT_CODE
1132445	Walker	32145	231
1217782	Adares	32145	125
1312243	Rakowski	34129	167
1321242	Rodriguez	37134	125
1542311	Smithson	37134	421
1657399	Vanloo	32145	231

Table name: AGENT

AGENT_CODE	AGENT_PHONE
125	6152439887
167	6153426778
231	6152431124
333	9041234445

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Data Dictionary and the System Catalog

- Data dictionary
  - Description of all tables in the database created by the user and designer
- System catalog
  - System data dictionary that describes all objects within the database
- Homonyms and synonyms must be avoided to lessen confusion
  - Homonym: same name is used to label different attributes
  - Synonym: different names are used to describe the same attribute

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Relationships within the Relational Database (1 of 4)

- One-to-many (1:M)
  - Norm for relational databases
- One-to-one (1:1)
  - One entity can be related to only one other entity and vice versa
- Many-to-many (M:N)
  - Implemented by creating a new entity in 1:M relationships with the original entities
  - Composite entity (i.e., bridge or associative entity): helps avoid problems inherent to M:N relationships
    - Includes the primary keys of tables to be linked

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Relationships within the Relational Database (2 of 4)

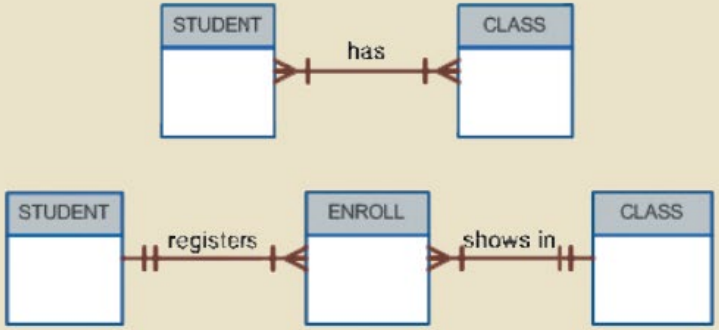
FIGURE 3.21 THE 1:1 RELATIONSHIP BETWEEN PROFESSOR AND DEPARTMENT



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Relationships within the Relational Database (3 of 4)

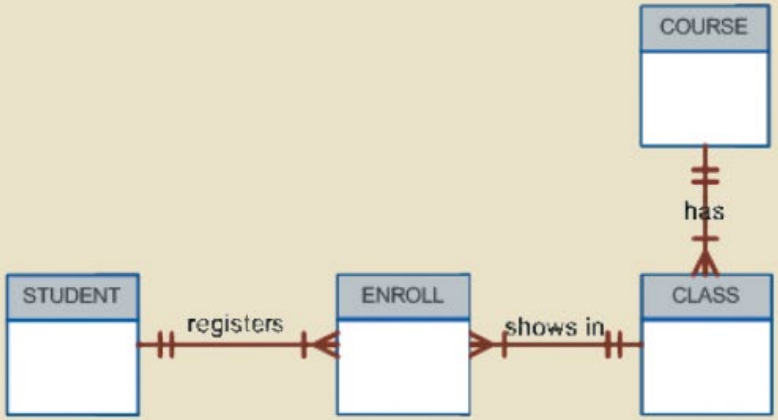
FIGURE 3.26 CHANGING THE M:N RELATIONSHIPS TO TWO 1:M RELATIONSHIPS



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Relationships within the Relational Database (4 of 4)

FIGURE 3.27 THE EXPANDED ER MODEL



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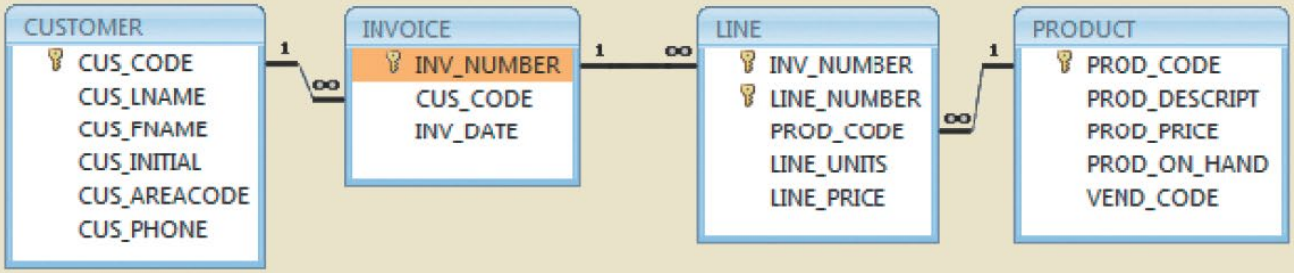
Data Redundancy Revisited (1 of 2)

- The relational database facilitates control of data redundancies through use of foreign keys
  - Common attributes that are shared by tables
- To be controlled except the following circumstances:
  - Sometimes data redundancy must be increased to make the database serve crucial information purposes
  - Sometimes data redundancy exists to preserve the historical accuracy of data

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Data Redundancy Revisited (2 of 2)

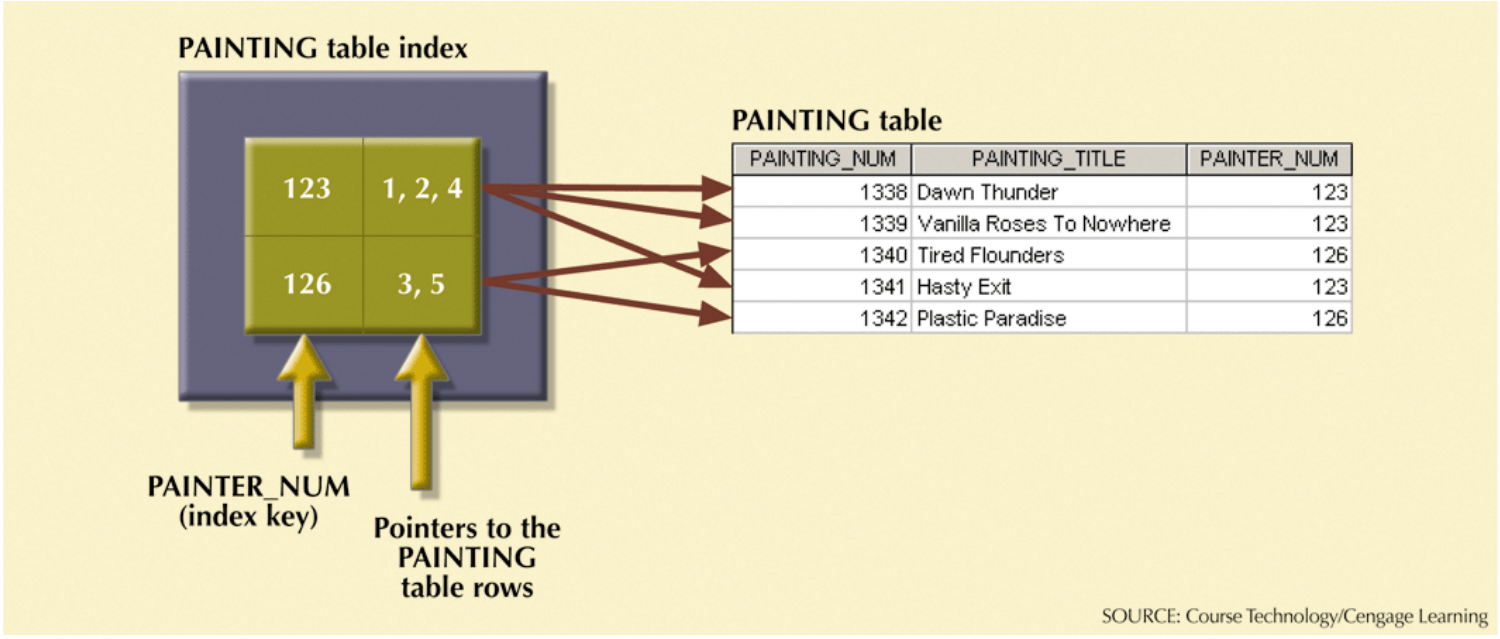
FIGURE 3.30 THE RELATIONAL DIAGRAM FOR THE INVOICING SYSTEM



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Indexes

- Orderly arrangement to logically access rows in a table
  - Index key: index's reference point that leads to data location identified by the key
  - Unique index: index key can have only one pointer value associated with it
- Each index is associated with only one table
  - The index key can have multiple attributes



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Codd's Relational Database Rules (1 of 2)

Table 13.8	Dr. Codd's 12 Relational Database Rules	
Rule	Rule Name	Description
1	Information	All information in a relational database must be logically represented as column values in rows within tables.
2	Guaranteed access	Every value in a table is guaranteed to be accessible through a combination of table name, primary key value, and column name.
3	Systematic treatment of nulls	Nulls must be represented and treated in a systematic way, independent of data type.
4	Dynamic online catalog based on the relational model	The metadata must be stored and managed as ordinary data—that is, in tables within the database; such data must be available to authorized users using the standard database relational language.
5	Comprehensive data sublanguage	The relational database may support many languages; however, it must support one well-defined, declarative language as well as data definition, view definition, data manipulation (interactive and by program), integrity constraints, authorization, and transaction management (begin, commit, and rollback).
6	View updating	Any view that is theoretically updatable must be updatable through the system.
7	High-level insert, update, and delete	The database must support set-level inserts, updates, and deletes.

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Codd's Relational Database Rules (2 of 2)

Table 13.8	Dr. Codd's 12 Relational Database Rules	
Rule	Rule Name	Description
8	Physical data independence	Application programs and ad hoc facilities are logically unaffected when physical access methods or storage structures are changed.
9	Logical data independence	Application programs and ad hoc facilities are logically unaffected when changes are made to the table structures that preserve the original table values (changing order of columns or inserting columns).
10	Integrity independence	All relational integrity constraints must be definable in the relational language and stored in the system catalog, not at the application level.
11	Distribution independence	The end users and application programs are unaware of and unaffected by the data location (distributed vs. local databases).
12	Nonsubversion	If the system supports low-level access to the data, users must not be allowed to bypass the integrity rules of the database.
13	Rule zero	All preceding rules are based on the notion that to be considered relational, a database must use its relational facilities exclusively for management.

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Summary

- Tables are the basic building blocks of a relational database
  - Keys are central to the use of relational tables
    - Each table row must have a primary key
  - Although tables are independent, they can be linked by common attributes
- The relational model supports several relational algebra functions
  - A relational database performs much of the data manipulation work behind the scenes
  - Once you know the basics of relational databases, you can concentrate on design

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